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2、

(a) ∵ $e^{ix} = \sum_{n=-\infty}^{\infty} J_{n(x)} i^{n}$,

⇒ $\cos x = J_{0(x)}(-J_{2(x)}+J_{4(x)}-\cdots)+(-J_{2(x)}+J_{4(x)}-\cdots)$

$= J_{0(x)}-2J_{2(x)}+2J_{4(x)}-2J_{6(x)}+\cdots$

$\sin x = 2[J_{1(x)}-J_{3(x)}+J_{5(x)}-J_{7(x)}+\cdots]$

得证

(b) $1 = J_0(0) = \sum_{k=-\infty}^{+\infty} J_{k(x)} J_{-k}(-x) = 2\sum_{k=1}^{+\infty} J_{k(x)}^2 + J_{0(x)}^2 = J_0^2 + 2[J_{1(x)}^2 + J_{2(x)}^2 + \cdots]$

3、

(a) $\int_0^x x^{-1} J_{4(x)} dx = \int_0^x x^2 x^{-3} J_{4(x)} dx = x^{-1} J_{3(x)}\Big|_0^x - 2x^{-2} J_{2(x)}\Big|_0^x = \frac{J_{3(x)}}{x} - 2\frac{J_{2(x)}}{x^2}$

$\int_0^x x^3 J_{0(x)} dx = \int_0^x x^2 \cdot x J_{0(x)} dx = x^3 J_{1(x)}\Big|_0^x - 2x^2 J_{1(x)}\Big|_0^x + 2x J_{1(x)}\Big|_0^x = x[x^2-2x+2] J_{1(x)}$

(b) $\int_0^x \cos x J_{0(x)} dx = x\cos x J_{0(x)} - xJ_1 \cos x$

令 $I_n = \int_0^x x^n J_{n(x)} \cos x dx = x^{n+1} J_{n(x)} \cos x - \int_0^x [x^{n+1} J_{n-1} \cos x - x^{n+1} J_n \sin x] dx$

$= \int_0^x (\frac{d}{dx}(x^{n+1} J_{n+1}) \sin x dx - (2n x^n \cdot J_n \cos x - x^{n+1} J_{n+1} \cos x) dx$

$= x^{n+1} J_{n(x)} \cos x + x^{n+1} J_{n+1} \sin x - I_{n+1} - 2n I_n + I_{n+1}$

⇒ $I_n = \frac{J_{n(x)}+J_{n+1(x)}}{2n+1} x^{n+1} \cos x$

6、

取平面极坐标系，其与φ无关，

$$\frac{d^2u}{dt^2} - \frac{c^2}{r}\frac{d}{dr}\left(r\frac{du}{dr}\right) = 0$$

$$u|_{t=0} = A\left(1-\frac{r^2}{R^2}\right), \quad \frac{du}{dt}\Big|_{t=0} = 0$$

$$u|_{r=0}\text{有界}, \quad u|_{r=R} = 0$$

其一般解：

$$u|_{r,t} = \sum_{i=1}^{\infty} C_i J_0\left(\mu_i \frac{r}{a}\right) e^{-c^2\left(\frac{\mu_i}{a}\right)^2 t}$$

代入初调，

$$u|_{t=0} = \sum_{i=1}^{\infty} C_i J_0\left(\mu_i \frac{r}{a}\right) = A\left(1-\frac{r^2}{R^2}\right)$$

$$\Rightarrow \quad u|_{r,t} = \sum_{i=1}^{\infty} \frac{8A}{\mu_i^3 J_1(\mu_i)} J_0\left(\mu_i \frac{r}{a}\right) e^{-c^2\frac{\mu_i^2}{a^2}t}$$

7、写得其一般解：

$$u = \sum_{i=1}^{\infty} C_i J_2\left(\mu_i^{(2)} \frac{r}{a}\right) \sin 2\varphi \, e^{-\frac{\mu_i^{(2)}}{a}kt}$$

代入初值得：

$$u = \sum_{i=1}^{\infty} \frac{-2\left(a\mu_i J_1(\mu_i a) + 2J_2(\mu_i a)\right)}{a^2 J_2'(\mu_i)^2 \mu_i^2} J_2\left(\mu_i^{(2)} \frac{r}{a}\right) \sin 2\varphi \, e^{-\frac{\mu_i}{a}kt}$$